

Model Context Protocol



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Agenda

What you can expect



1. Basics and Motivation
2. MCP-Architecture
3. MCP Pillars
4. Use existing MCP servers
5. Develop a custom MCP server
 - a) Local files
 - b) Taking screenshots
 - c) Making API calls
6. MCP Server Deployment



Basics and Motivation

Basics and Motivation

Intuition: File Formats and conversion

Problems

(1) LLMs cannot interact with outside world

(2) Function calling / tools overcome (1), but mechanisms are not standardized

(3) Tools often run on servers, thus cannot operate on local machine

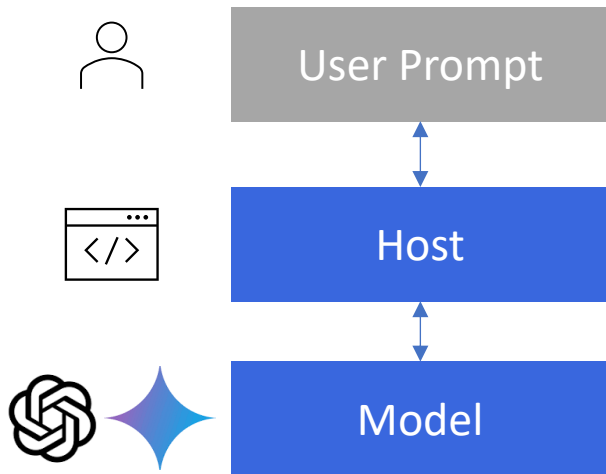
Need for standardized approach so that LLMs can interact with external systems



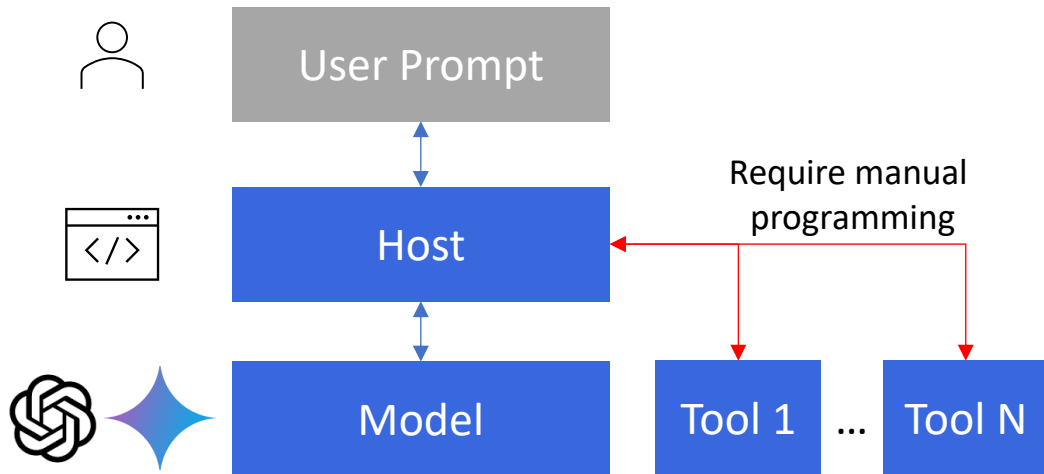
Basics and Motivation

Traditional Tool Use

Without Tools

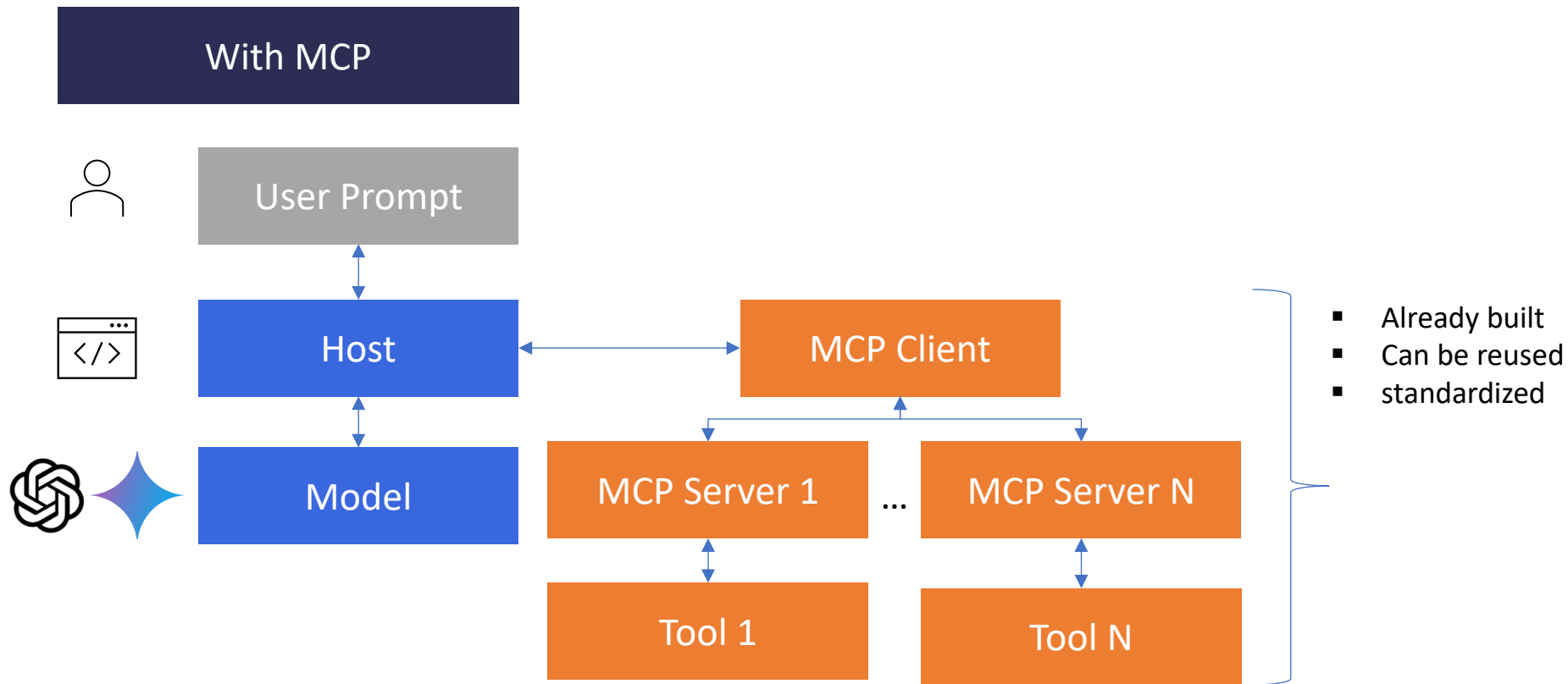


With Tools



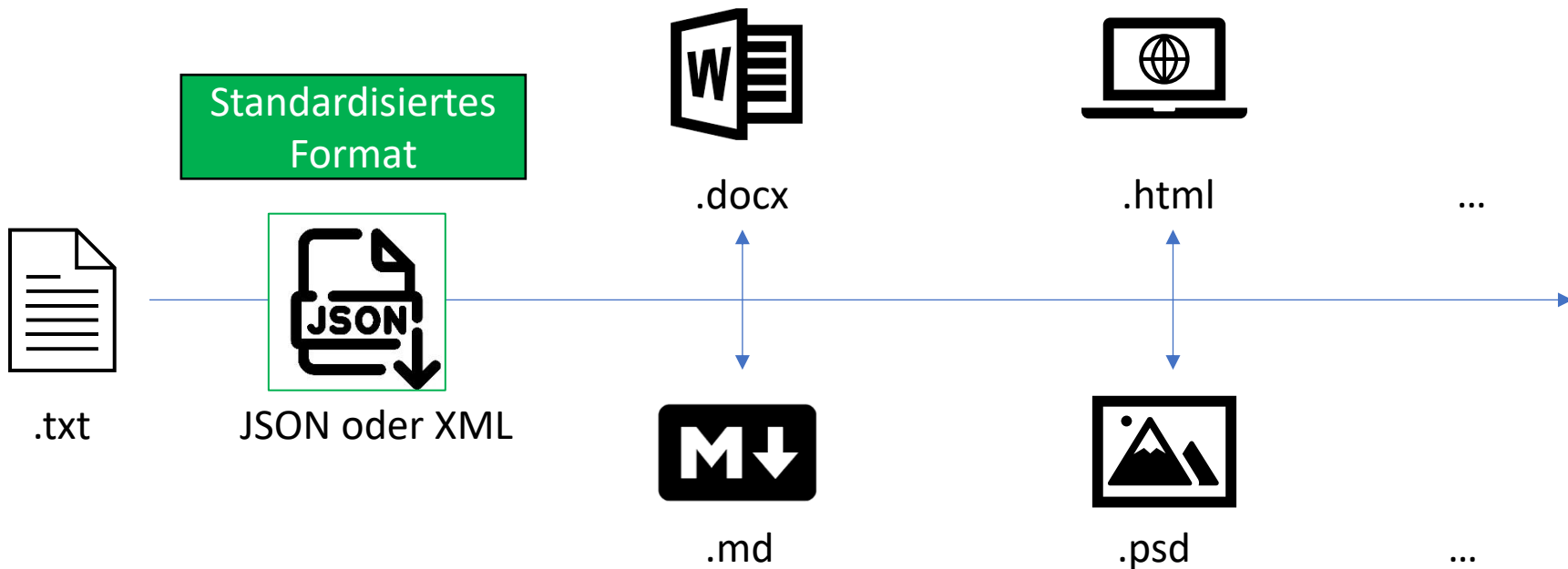
Basics and Motivation

Tool Use with MCP



Basics and Motivation

Intuition: File Formats and conversion

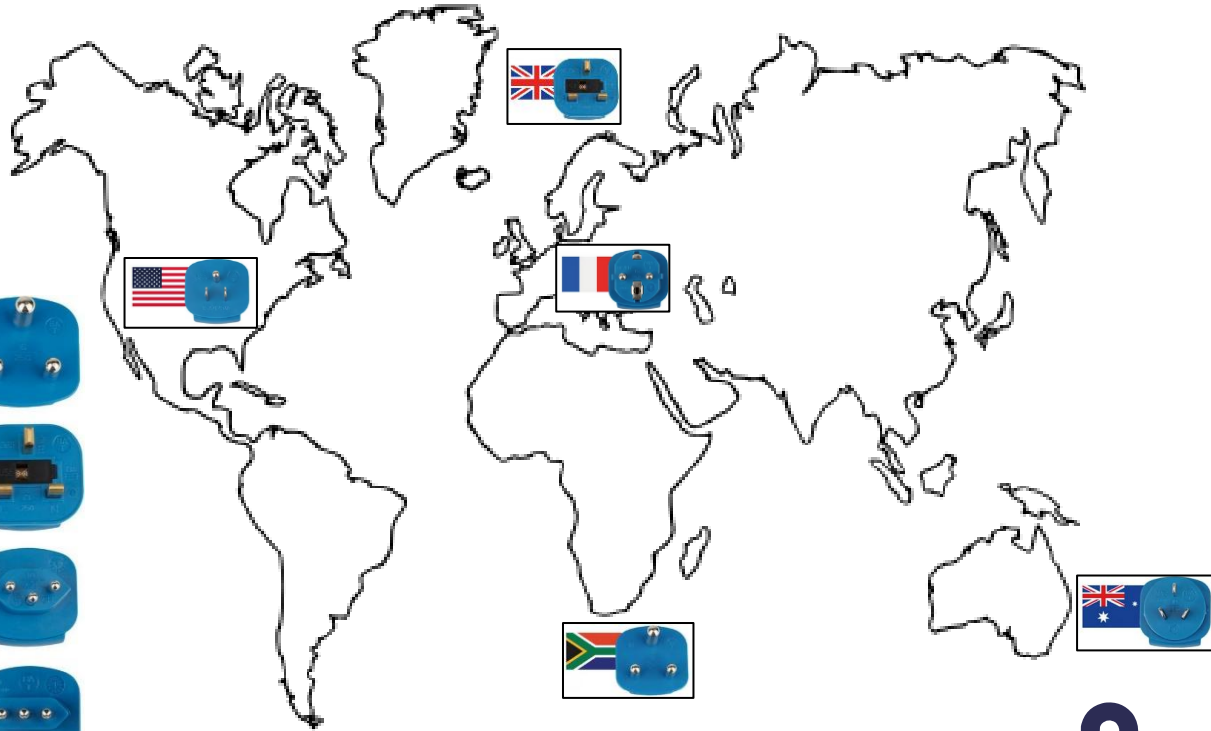


Basics and Motivation

Intuition: power adapters

Lösung:

Universalstecker



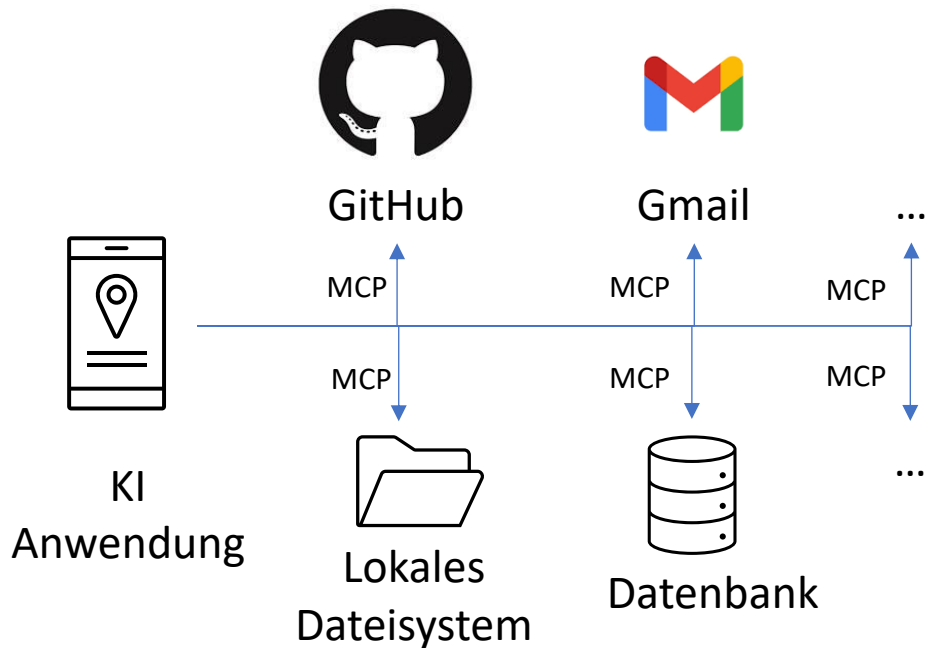
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Basics and Motivation

Intuition: without MCP

World before MCP

- A connection from each external tool has to be developed from scratch
- Each developer has to redevelop the wheel.

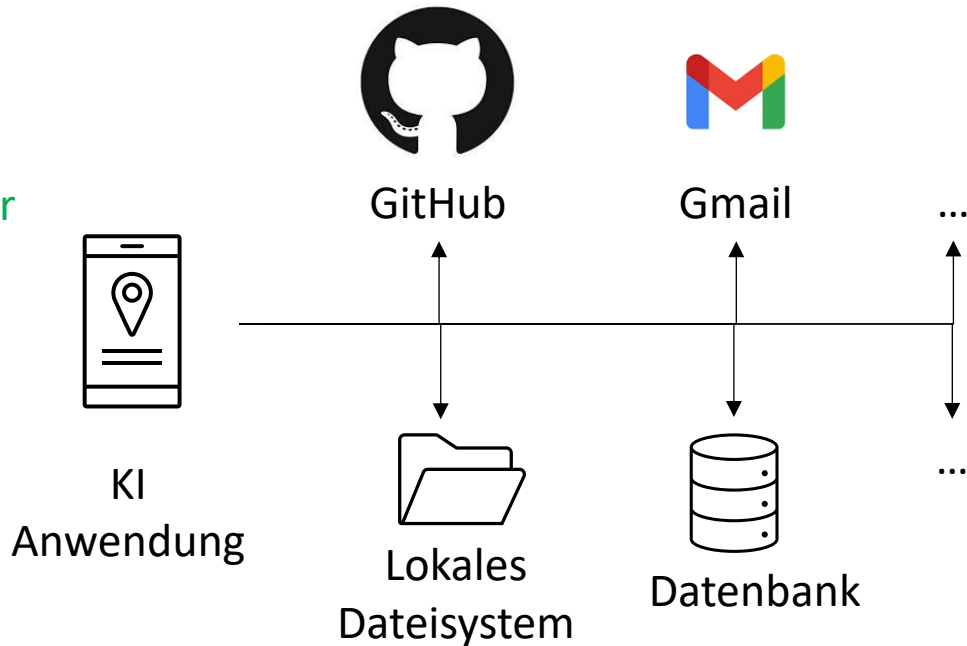


Basics and Motivation

Intuition: Ohne MCP

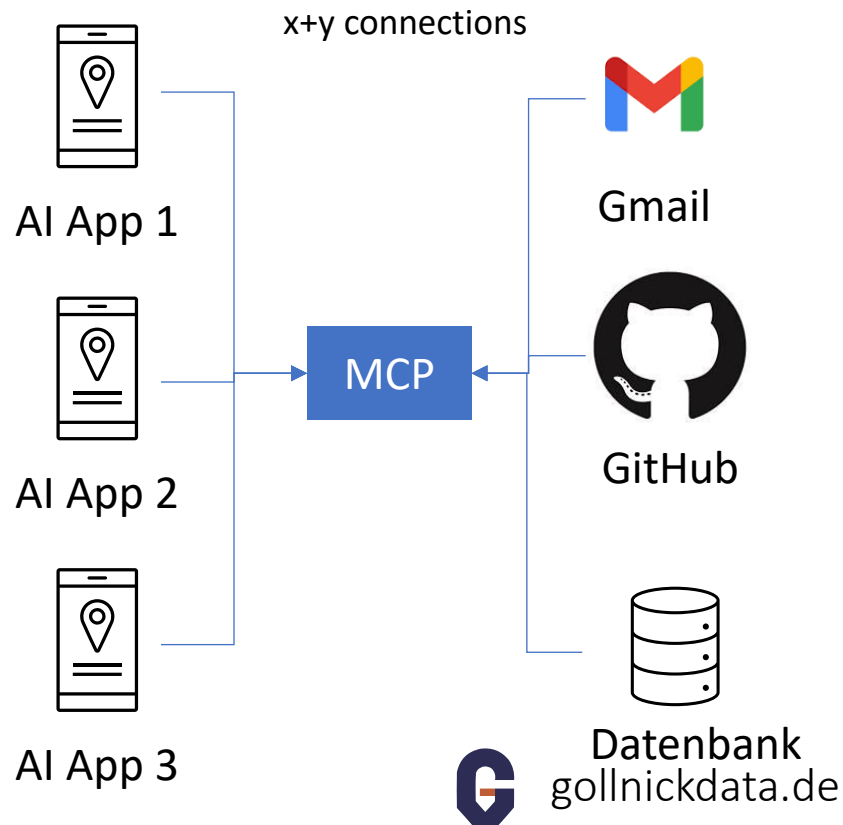
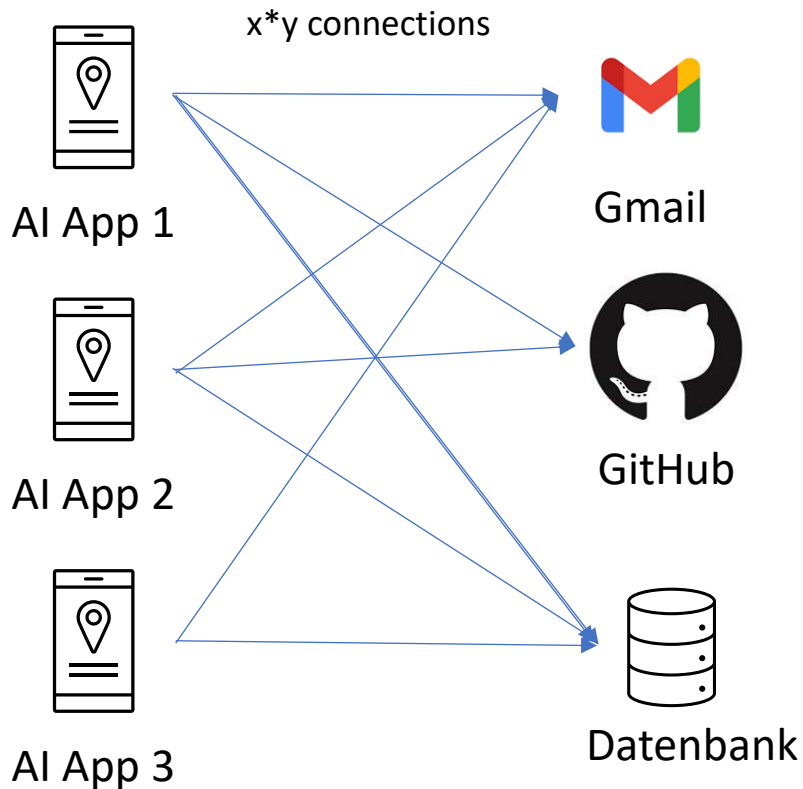
World with MCP

- A single connection to MCP server needs to be established.
- Each developer uses the same pre-developed connections.



Basics and Motivation

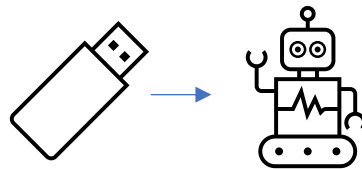
Intuition: Comparison with and without MCP



Basics and Motivation

What is MCP?

- Open-standard
- Open-source framework
- Connects agents to tools, APIs
- Uses structured inputs/outputs
- Introduced by Anthropic in 11/2024
- Standardize how AI agents
 - Integrate data
 - Share data with external tools



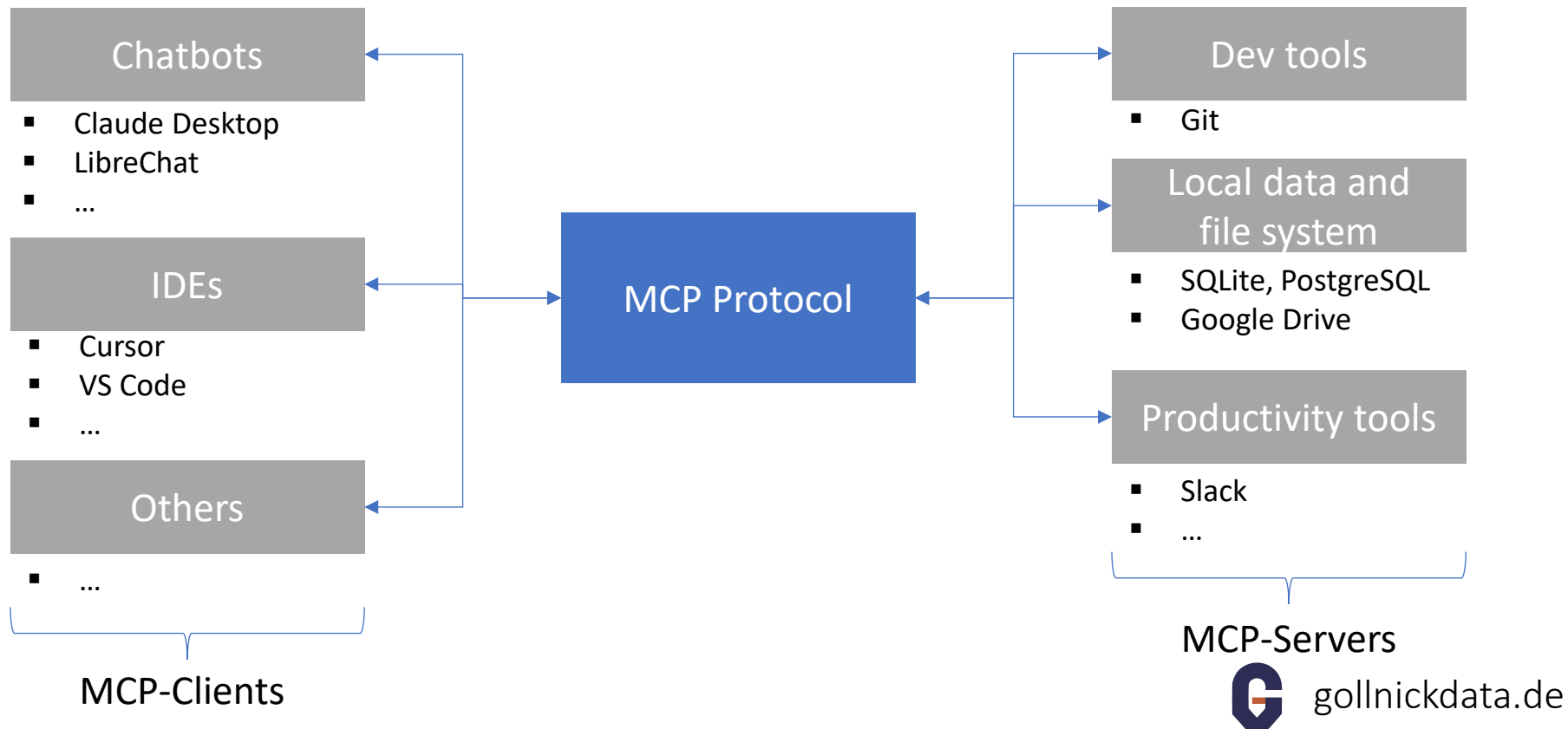
USB for AI



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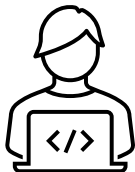
Basics and Motivation

Clients and Servers



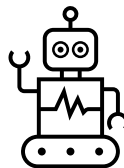
Basics and Motivation

Benefits of MCP



Developers

- Development time ↓
- Code complexity ↓



AI apps / agents

- Access to ecosystem of data sources, tools and apps
- Capabilities ↑
- End-user experience ↑



End User

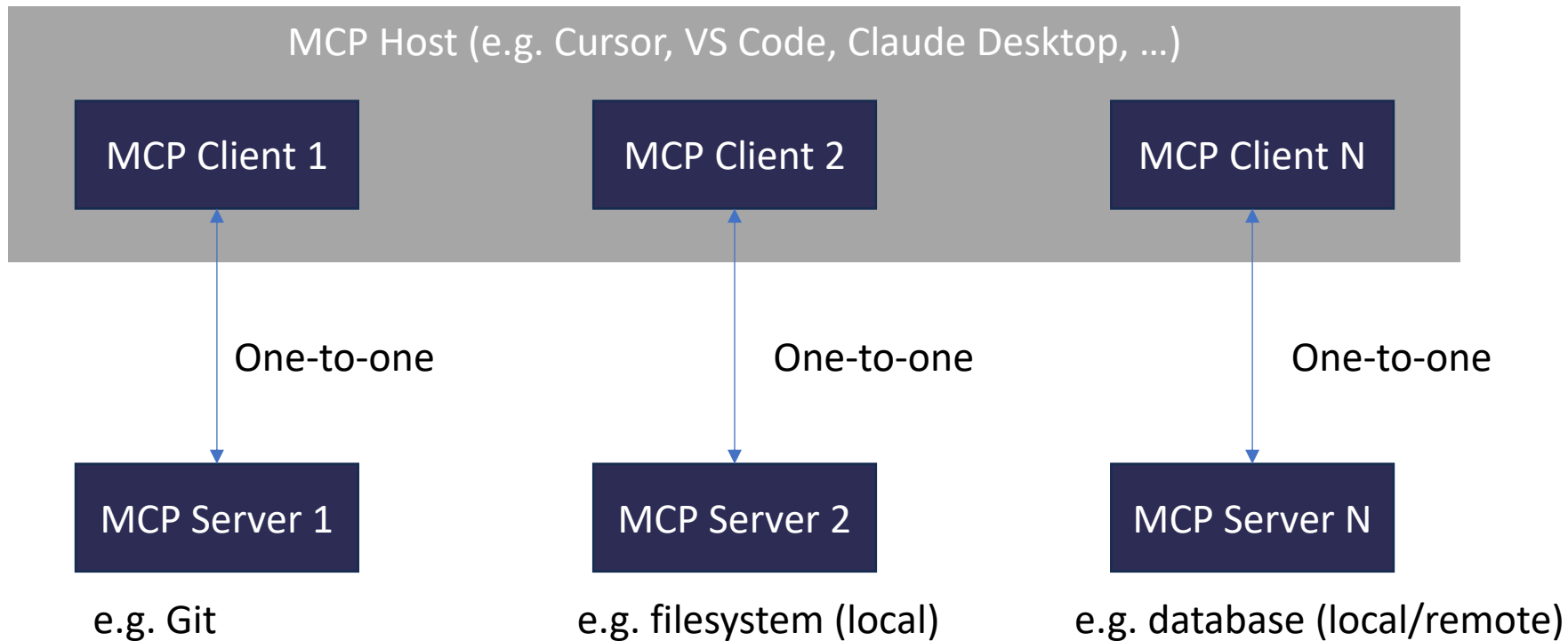
- More capable AI apps or agents
- Access to data
- Agents can take actions on behalf of user



MCP-Architecture

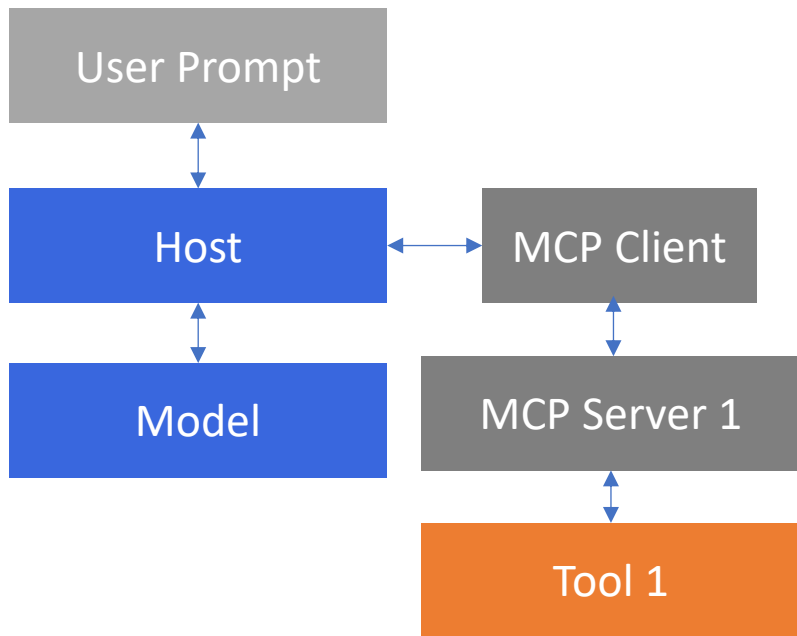
MCP-Architecture

Host, Client, and Server



MCP-Architecture

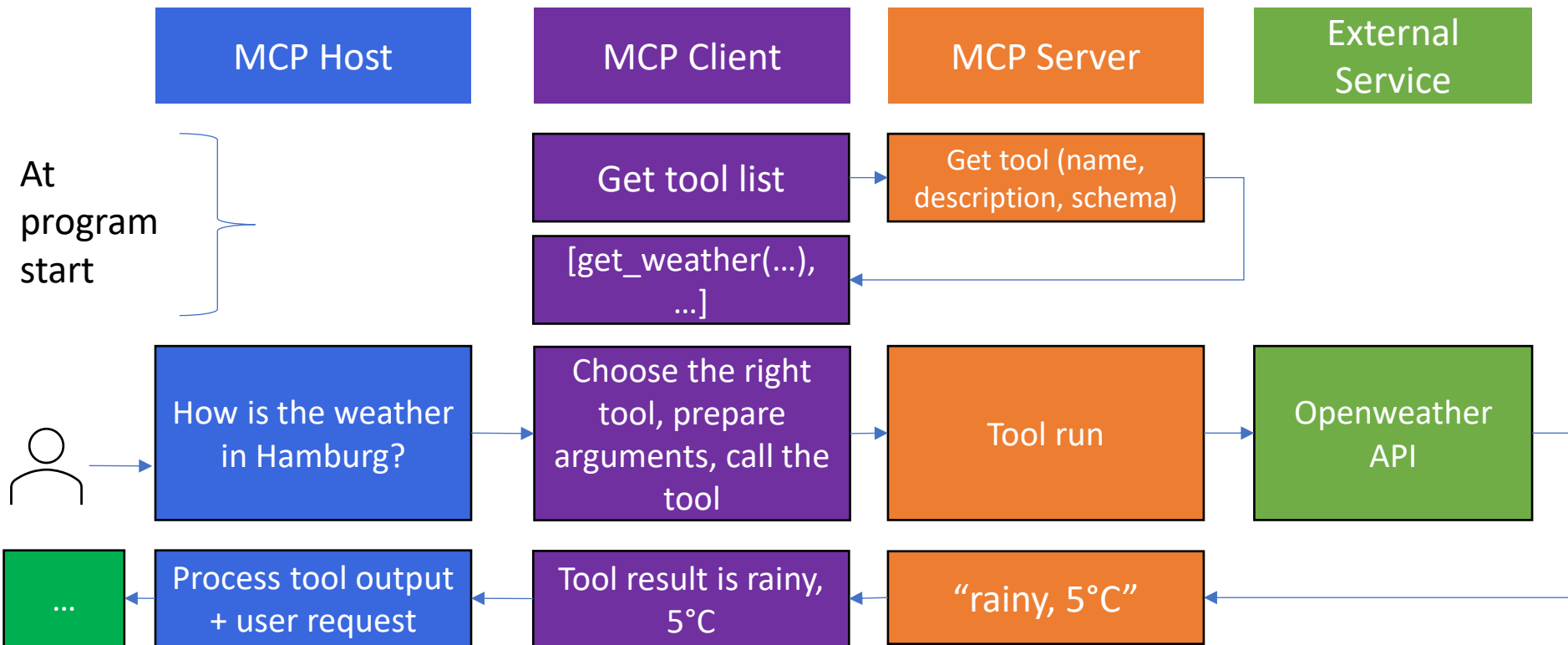
Tool Definition



Parameter	Value	Example
Name	Tool or function name	„get date“, „get weather“, ...
Description	What does the tool or function do, which parameters are expected	„get the weather for a given location, location as lat long“
Input Schema	Dictionary of Parameters that the tool accepts	{ „lat“: 14, „long“: 13 }

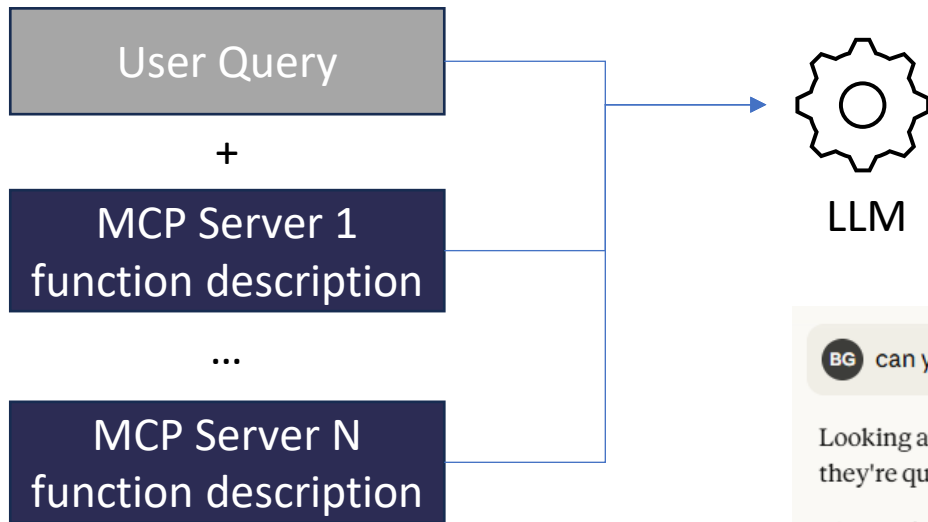
MCP-Architecture

Communication



MCP-Architecture

Challenges: Token Overhead



BG can you check how many tokens are sent for the installed mcp servers

Looking at the MCP server function descriptions that are sent with each request, I can see they're quite detailed. Let me give you a rough estimate:

The **arXiv MCP Server** functions are relatively concise - probably around **800-1,200 tokens** total for all 4 functions.

The **Blender MCP Server** is much more extensive with about 20 functions, and many include detailed descriptions and usage guidelines. This is likely around **3,000-4,000 tokens** total.

So combined, the MCP server function descriptions are probably consuming approximately **4,000-5,000 tokens** of context with each request.

MCP Pillars

MCP Pillars

MCP Server Components

Tools

- Functions developed
- Can have API requests, calculations, database actions

Resources

- Contextual data for host
- File contents
- User data

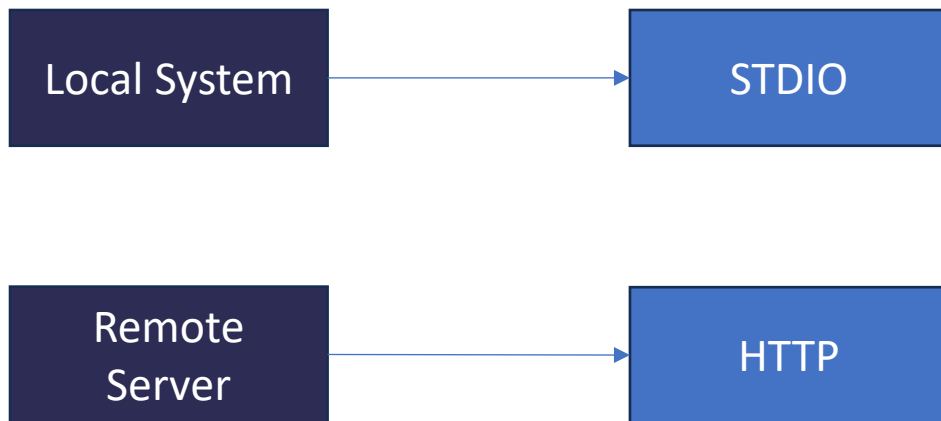
Prompts

- User-controlled prompt templates



MCP Pillars

Local vs. Remote Server



Use existing MCP servers

Use existing MCP servers

MCP Clients

 Continue by Continue Dev, Inc.	 Claude Desktop by Anthropic	 Zed by Zed Industries	 Sourcegraph Cody by Sourcegraph
Coding assistant plugin for VSCode and JetBrains. Highly configurable and open source.	Chat with Claude. Anthropic's native implementation for macOS and Windows.	Code editor. Highly performant and open source.	Coding assistant plugin for VSCode and JetBrains.
 Cline by Cline Bot Inc.	 ChatMCP by daodao97	 HyperChat by Big Sweet Potato Studio	 Sage by Michael Latman
Coding assistant plugin for VSCode with autonomous CLI capabilities. Open source.	Chat with LLM's. UI for macOS, Windows, Linux.	Chat with LLM's for macOS and Windows. Open Source.	Chat with Claude for macOS, iOS, and iPadOS.
 LibreChat by Danny Avila	 Sire by UOMS	 Systemprompt by Ejb503	 Toolbase by Toolbase
Chat with LLM's for macOS and Windows. Open source.	Chat with LLM's for macOS and Windows. Open source.	Gemini-powered MCP server control with your voice.	Enhance Claude Desktop with baked-in, one-click toggle MCP capabilities. Open source.
 CLI Inspector by wong2	 Roo Code (prev. Roo Cline) by Roo Vet	 Goose by Block, Inc.	 Cursor IDE by Anysphere
CLI tool for interacting directly with MCP servers via stdio or SSE, enabling fine-grained exploration of Tools, Resources, and Prompts.	Fork of Cline. Coding assistant plugin for VSCode with autonomous CLI capabilities. Open source.	Configurable software engineering AI agent that runs on your local machine.	AI-powered code editor

Source: <https://www.pulsemcp.com/clients>



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Use existing MCP servers

MCP Servers

 ArXiv John Blazick • Nov 29, 2024 Search and analyze academic papers from the arXiv repository. ↓ 65.6k ☆ 200	 MCP DB Utils Dong Hao • Feb 10, 2025 DButils is an all-in-one MCP service that enables your AI to do data analysis by accessing versatile types of database (sqlite, mysql, postgres, and more) within a unified connection configuration in a safe way. ↓ 54.6k ☆ 41	 arXiv Research Andy Brandt • Dec 7, 2024 Search and analyze scientific papers from arXiv. ↓ 52.2k ☆ 38
 LLM Code Context cyberchitta • Dec 6, 2024 Streamlines code context sharing with LLMs by implementing smart file selection, code outlining, and multi-language support for efficient code reviews and documentation generation. ↓ 49.6k ☆ 153	 Crawl4AI (Web Scraping & Crawling) Ritvij Kumar Sharma • Mar 6, 2025 Integrates web scraping and crawling capabilities with tools for single-page content extraction, metadata retrieval, and Google search functionality, enabling workflows that require web content analysis, data collection, or online research tasks. ↓ 47.8k ☆ 1	 Cloudflare Cloudflare • Nov 27, 2024 Manage KV stores, R2 storage, D1 databases, and Workers deployments. ↓ 43.6k ☆ 1k
 DuckDuckGo Search Nick Clyde • Feb 23, 2025 Integrates with DuckDuckGo to provide web search capabilities, content fetching, and parsing, with results formatted for large language model consumption. ↓ 37.3k ☆ 42	 KOSPI/KOSDAQ Stock Data Sangrok Mun • Feb 14, 2025 Integrates with pykrx to provide real-time access to KOSPI and KOSDAQ stock market data, offering tools for retrieving ticker symbols, OHLCV data, market capitalization, fundamental metrics, and trading volumes. ↓ 35.3k ☆ 16	 Sentry Anthropic • Nov 19, 2024 Retrieve and analyze application error reports for debugging support. ↓ 35.1k ☆ 26k

Source: <https://www.pulsemcp.com/servers>



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AirBNB

Use existing MCP servers

Find MCP Servers

Live Demo



Search

1. search

Airbnb

OpenBnB

Integrates with Airbnb to enable vacation rental search and detailed property information retrieval without requiring...

CLASSIFICATION

 Community

EST VISITORS

(WEEK)

1.1k

RELEASE DATE

Mar 10, 2025

2. select



GitHub Repo (359 stars)

3. Read details



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Use existing MCP servers

Install MCP Servers

For Claude Desktop

This extension is packaged as a Desktop Extension (DXT) file. To install:

1. Download the `.dxt` file from the releases page
2. Open your compatible AI application (e.g., Claude Desktop)
3. Install the extension through the application's extension manager
4. Configure the extension settings as needed

Releases 1

 **v0.1.3** Latest
on Jul 18, 2025

<https://github.com/openbnb-org/mcp-server-airbnb/releases>

Download Extension file (Dxt)

For Cursor, etc.

Before starting make sure [Node.js](#) is installed on your desktop for `npx` to work.

1. Go to: Cursor Settings > Tools & Integrations > New MCP Server
2. Add one the following to your `mcp.json` :

```
{
  "mcpServers": {
    "airbnb": {
      "command": "npx",
      "args": [
        "-y",
        "@openbnb/mcp-server-airbnb"
      ]
    }
  }
}
```

or

Live Demo



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Use existing MCP servers

Install MCP Servers: Claude Desktop Extension

Live Demo

Einstellungen

Allgemein

Konto

Datenschutz

Abrechnung

Fähigkeiten

Konnektoren

Claude Code

Desktop-App

Allgemein

Erweiterungen

Entwickler

Erweiterungsentwickler



Entwicklertools-Warnung

Diese Tools sind ausschließlich für Erweiterungsentwickler gedacht. Eine falsche Verwendung kann zu Fehlfunktionen von Erweiterungen führen oder die Sicherheit deines Systems gefährden.

Erweiterung installieren

Entpackte Erweiterung installieren

Erweiterungsordner öffnen

Erweiterungseinstellungen-Ordner öffnen

< Vorschau schließen

A

Airbnb Search & Listings

Search Airbnb listings with advanced filtering and get detailed property information

Installieren



Die Installation gewährt Zugriff auf alles auf deinem Computer.

Use existing MCP servers

Activate MCP Servers

Live Demo

Wie kann ich dir heute helfen?



Dateien oder Fotos hinzufügen



Zum Projekt hinzufügen



Von Airtable MCP Server hinzufü...



Websuche



Stil verwenden



Konnektoren



airbnb



Airbnb Search & Listings



Airtable MCP Server



LocalMcpServer



n8n-mcp



n8n-workflows Docs



Windows-MCP



Konnektoren verwalten



Activate MCP servers
to be used



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Use existing MCP servers

Test MCP Servers

welche unterkünfte gibt es im Harz, konkret Bad Lauterberg und 10km Entfernung. Sauna ist Must have. Sterne: mindestens 4.5

Live Demo

A Claude möchte Airbnb search von Airbnb Search & Listings verwenden

Anfrage

```
{  
  `location`: `Bad Lauterberg, Harz, Germany`  
}
```

Ablehnen Esc

Immer erlauben Ctrl Enter

Einmal erlauben Enter



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Use existing MCP servers

Test MCP Servers

Live Demo

welche unterkünfte gibt es im Harz, konkret Bad Lauterberg und 10km Entfernung. Sauna ist Must have. Sterne: mindestens 4.5

Ergebnis der Suche

Leider gibt es auf Airbnb in Bad Lauterberg und Umgebung **nur sehr wenige Unterkünfte mit privater Sauna**. Von den über 20 geprüften Optionen mit 4,5+ Sternen habe ich nur **eine Unterkunft mit Sauna-Zugang** gefunden:

★ MeLou Fewo Harzruhe Wellness

- **Rating:** 4.71/5 (14 Bewertungen)
- **Ausstattung:** Shared Sauna (kostenlos Do/Fr/Sa/So 16-22 Uhr, sonst gegen Kleingeld), Indoor Pool 24h, Balkon mit Bergblick
- **Größe:** ca. 30 m², 1-2 Personen
- **Preis:** ab €341/5 Nächte
- **Link:** <https://www.airbnb.com/rooms/1470033092657220820>
- **Besonderheit:** Hundefreundlich, direkt am Waldrand, 15 Min. zu Fuß zum Zentrum



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arXiv articles

Projects and Applications

Arxiv und Claude Desktop: Setup

arXiv MCP Server

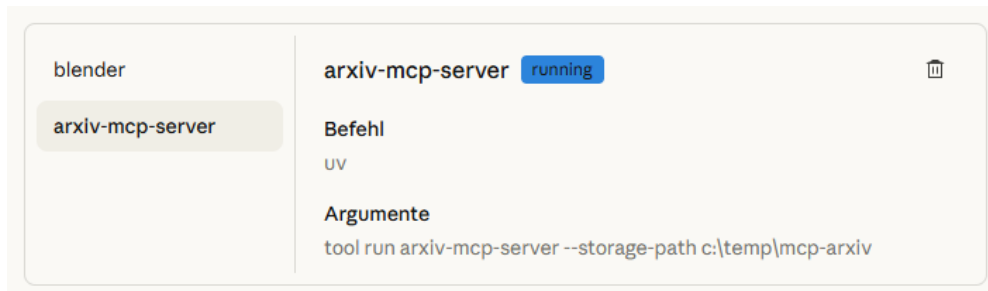
Search and analyze academic papers on arXiv.

Installation:

uv tool install arxiv-mcp-server

```
"arxiv-mcp-server": {  
  "command": "uv",  
  "args": [  
    "tool",  
    "run",  
    "arxiv-mcp-server",  
    "--storage-path",  
    "c:\\\\temp\\\\mcp-arxiv"  
  ]  
}
```

← Adapt storage path



Arxiv MCP

<https://mcpservers.org/servers/github-com-blazickjp-arxiv-mcp-server>

Claude



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Projects and Applications

Arxiv und Claude Desktop: Use

arXiv MCP Server

Search and analyze academic papers on arXiv.

Claude

 suche nach llm papern auf arxiv mit deinem mcp server

Ich suche nach aktuellen LLM-Papers auf arXiv für dich.

 Search papers 

Hier sind 20 aktuelle LLM-Papers von arXiv (sortiert nach neuesten zuerst):

Herausragende Papers:

- 1. The Universal Weight Subspace Hypothesis** (2512.05117) Zeigt, dass neuronale Netze über verschiedene Aufgaben hinweg bemerkenswert ähnliche niedrigdimensionale Unterräume aufweisen. Analyse von über 1100 Modellen (500 Mistral-7B LoRAs, 500 Vision Transformers, 50 LLaMA-8B) identifiziert universelle Unterräume.
- 2. Semantic Soft Bootstrapping** (2512.05105) Eine Alternative zu Reinforcement Learning für Long-Context-Reasoning. Self-Distillation-Technik ohne RL, die auf GSM8K 10,6% bessere Genauigkeit als GRPO erreicht.
- 3. TV2TV: Interleaved Language and Video Generation** (2512.05103) Vereint Video- und Text-Generation durch verschachtelte Erzeugung - das Modell kann "in Worten denken" bevor es "in Pixeln handelt".



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Projects and Applications

Arxiv und Claude Desktop: Use

arXiv MCP Server

Search and analyze academic papers on arXiv.



C:\\temp\\mcp-arxiv



2512.05117.pdf



2512.05117.md

Claude



lade mir das erste runter



Download paper



Das Paper wird gerade heruntergeladen und konvertiert. Lass mich den Status überprüfen:



Download paper



Die Konvertierung läuft noch. Lass mich kurz warten und dann nochmal checken:



Download paper



Die Konvertierung dauert noch einen Moment. Lass mich einen weiteren Check machen:



3 Schritte



Download paper



Download paper



Perfekt! Das Paper wurde erfolgreich heruntergeladen und konvertiert. Lass mich es jetzt für dich lesen:

Projects and Applications

Arxiv und Claude Desktop: Use

arXiv MCP Server

Search and analyze academic papers on arXiv.



C:\\temp\\mcp-arxiv



2512.05117.pdf



2512.05117.md

Claude

Perfekt! Ich habe das Paper "**The Universal Weight Subspace Hypothesis**" erfolgreich heruntergeladen und gelesen. Hier ist eine Zusammenfassung der wichtigsten Punkte:

Kernaussage

Das Paper zeigt, dass neuronale Netze, die über verschiedenste Aufgaben trainiert werden, **bemerkenswert ähnliche niedrigdimensionale Parameterunterräume** aufweisen - unabhängig von Initialisierung, Task oder Domäne.

Hauptergebnisse

Empirische Evidenz:

- Analyse von **über 1100 Modellen**: 500 Mistral-7B LoRAs, 500 Vision Transformers, 50 LLaMA-8B
- Identifizierung **universeller Unterräume**, die die Mehrheit der Varianz in nur wenigen Hauptrichtungen erfassen (oft 16 oder weniger)

Theoretische Grundlage:

- Beweis der Konvergenz mit Rate $O(1/\sqrt{T})$
- Two-level convergence theorem für geteilte Unterräume
- Zeigt, dass die Qualität der Approximation r^2 mit der Anzahl der Modelle zunimmt



3D Objects with Blender

Projects and Applications

3D Objects in Blender with LLM

3d modell eines hauses in blender erstell

das dach sollte ein klassisches spitzdach sein. bitte passe es direkt an

Ausführen des Skripts, um das Haus mit dem neuen Spitzdach zu erstellen:

```
Auto-Run command: python  
$ python create_house_example.py  
Scale: 1.0  
C:\Temp\blender-open-mcp>
```

Das Haus wurde erfolgreich mit einem klassischen Spitzdach erstellt.
Das Dach:

- Zwei geneigte Seiten, die sich in der Mitte treffen
- Einen First (obere Kante)
- Eine höhere Spitze als das flache Dach

Der Code wurde in beiden Dateien angepasst:

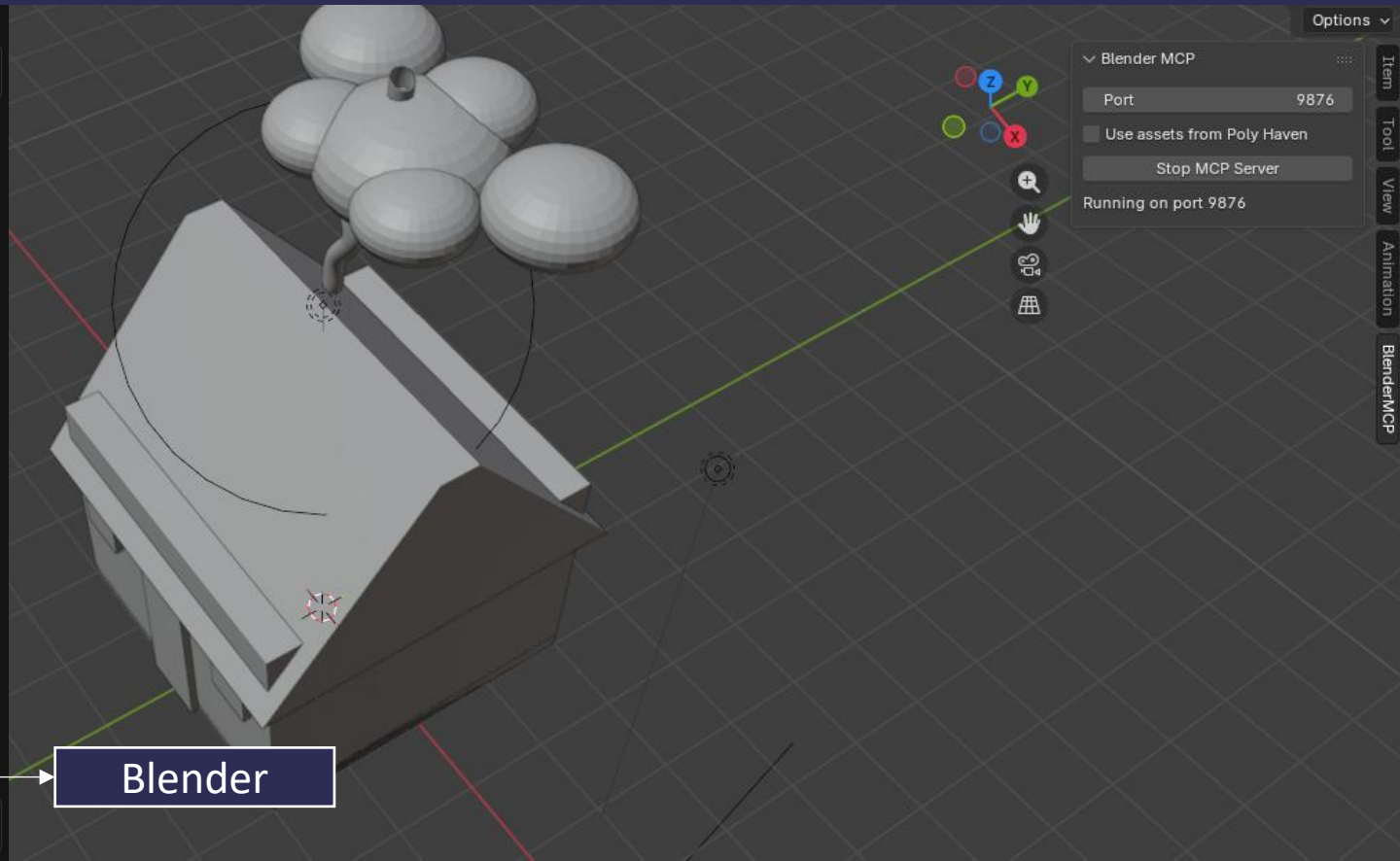
- `create_house_example.py` – für direkte Ausführung
- `src/blender_open_mcp/server.py` – für das MCP-Tool

Das Spitzdach wird erstellt, indem:

1. Ein Cube als Basis erstellt wird
2. Die oberen Vertices ausgewählt werden
3. Diese nach oben verschoben werden, um die Spitze zu bilden
4. Die Vertices entlang der Y-Achse skaliert werden, um den First zu formen

Das Haus sollte jetzt in Blender mit dem neuen Spitzdach sichtbar sein.

super. eine sonne soll auf das haus scheinen und eine wolke über dem haus schweben



Projects and Applications

3D Objects in Blender with LLM: MCP Server Setup

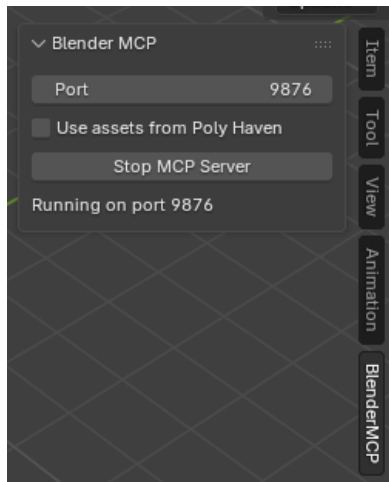
MCP Server finden

<https://www.pulsemcp.com/servers/dhakalnirajan-blender>

Server installieren

<https://github.com/dhakalnirajan/blender-open-mcp>

Server starten



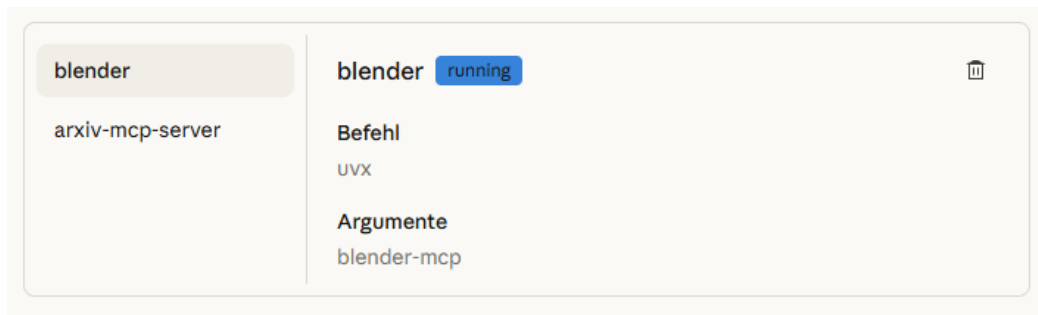
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Projects and Applications

3D Objects in Blender with LLM: MCP Client Setup

Claude Desktop Integration

```
{ "mcpServers": {  
  "blender": {  
    "command": "uvx",  
    "args": [  
      "blender-mcp"  
    ]  
  }  
}
```



Projects and Applications

3D Objects in Blender with LLM: Use

BG create a simple house in blender via mcp

I'll help you create a simple house in Blender using the MCP tools. Let me start by checking the current scene and then build a basic house structure.

B Get scene info

Good! Now let me create a simple house with walls, a roof, a door, and some windows. I'll build it step by step:

^ 3 Schritte

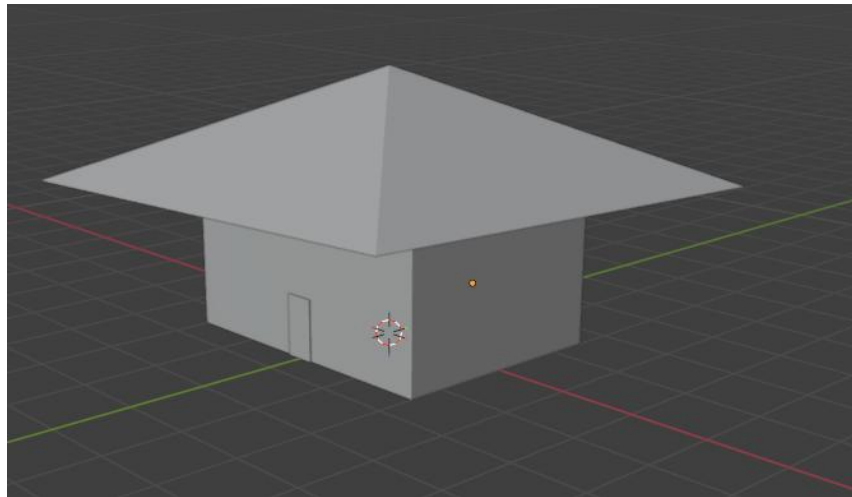
B Execute blender code

B Execute blender code

Perfect! Now let me take a screenshot to show you the house:

B Get viewport screenshot

Let me check the scene info to confirm everything was created successfully:



Claude

Blender



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Develop custom MCP-Server: Local Files

Custom MCP Server

MCP Servers

- Show slideset on environments
- Set up uv environment

Live Demo



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Custom MCP Server

MCP Servers

code > server.py >

```
from mcp.server.fastmcp import FastMCP
import os
### Create an MCP server
mcp = FastMCP("LocalMcpServer")
### Add an addition tool
@mcp.tool()
def list_files(folder_path: str) -> list[str]:
    """List files in a folder"""
    return os.listdir(folder_path)

@mcp.tool()
def write_markdown_file(file_path: str, content: str) -> str:
    """Write a markdown file
    Args:
        file_path: path to the file
        content: content to write to the file
    Returns:
        success message
    """
    with open(file_path, 'w') as file:
        file.write(content)
    return f"File {file_path} written successfully"
if __name__ == "__main__":
    mcp.run()
```

Live Demo



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Custom MCP Server

Test MCP Servers

Run MCP inspector

```
mcp dev server.py
```

▶ Connect

Live Demo



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Custom MCP Server

Test MCP Servers

Live Demo

Resources Prompts **Tools** Ping Sampling Elicitations Roots Auth Metadata

Tools



List Tools

Clear

list_files

List files in a folder



write_markdown_file

Write a markdown file Args: file_path: path to the file content: content to write to the file Returns: success message



list_files

List files in a folder

folder_path *

C:\Temp

Tool-specific Metadata:

No metadata pairs.

Add Pair

Output Schema:

Expand

```
{
  "result": []
}
title: "list_filesOutput"
```

Run Tool

Copy Input

Tool Result: **Success**

Custom MCP Server

Test MCP Servers

Live Demo

Tool Result: **Success**

Structured Content:

```
{
  result: [
    0: "01_Preface.zip"
    1: "02_Introduction_DeepLearning.zip"
    2: "AI-Learning"
    3: "AiEngineering"
    4: "AiEngineering_2025-11"
    5: "Alterric"
  ]
}
```

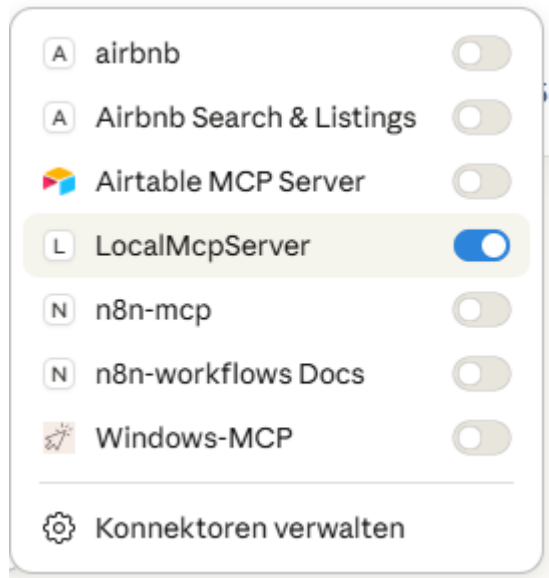


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Custom MCP Server

Install MCP Servers (in Claude Desktop)

```
mcp install server.py
```



welche dateien habe ich im unterordner temp

Live Demo



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Develop custom MCP-Server:
Take Screenshots

Custom MCP Server

MCP Servers

code >  screenshot.py >

```
# /// script
# requires-python = ">=3.12"
# dependencies = [
#     "mcp",
#     "pyautogui",
#     "pillow",
# ]
# ///
# %% packages
from mcp.server.fastmcp import FastMCP
from mcp.server.fastmcp.utilities.types import Image
import pyautogui
import io
mcp = FastMCP("Take Screenshot")
@mcp.tool()
def capture_screenshot() -> Image:
    """    Capture the current screen and return the image.    """
    buffer = io.BytesIO()
    screenshot = pyautogui.screenshot()
    screenshot.convert("RGB").save(buffer, format="JPEG", quality=60, optimize=True)
    return Image(data=buffer.getvalue(), format="jpeg")
if __name__ == "__main__":
    mcp.run()
```

Live Demo



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Custom MCP Server

Test MCP Servers

Run MCP inspector

```
mcp dev screenshot.py
```

▶ Connect

Live Demo



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Custom MCP Server

Test MCP Servers

Live Demo

Resources Prompts **Tools** Ping # Sampling Elicitations Roots Auth Metadata

Tools



List Tools

Clear

capture_screenshot



Capture the current screen and return the image. Use this tool whenever the user requests a screenshot of their activity.

capture_screenshot

Capture the current screen and return the image. Use this tool whenever the user requests a screenshot of their activity.

Tool-specific Metadata:

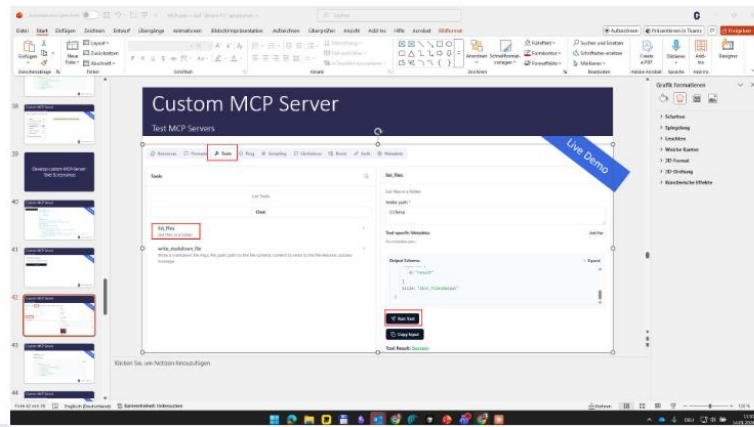
No metadata pairs.

Add Pair

Run Tool

Copy Input

Tool Result: **Success**



Custom MCP Server

Install MCP Servers (in Claude Desktop)

```
mcp install screenshot.py
```

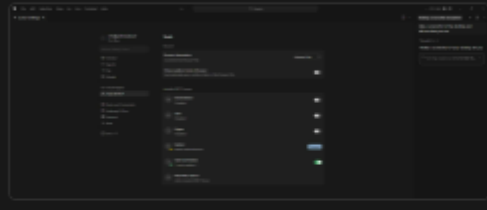
```
"Take Screenshot": {  
  "command": "uv",  
  "args": [  
    "run"  
    "C:\\\\Projects\\\\LiveTrainings\\\\MCP_Kurzkurs  
    \\\\code\\\\screenshot.py"  
  ]  
}
```

take a screenshot of my desktop and tell me
what you see.

Thought for 2s

I'll take a screenshot of your desktop for you.

✓ Ran capture_screenshot:Take Sc...



Thought for 2s

Here's what I can see on your desktop:

You have Cursor IDE open with the settings panel displayed. Specifically, you're viewing the "Tools & MCP" (Model Context Protocol) settings section. Here are the details:

Live Demo

Develop custom MCP-Server: Making API-Calls

Custom MCP Server: API-Calls

MCP Servers

code >  get_crypto_prices.py >

```
# /// script
# requires-python = ">=3.12"
# dependencies = [
#     "mcp",
#     "requests",
# ]
# ///

from mcp.server.fastmcp import FastMCP
import requests

mcp = FastMCP("Crypto")

@mcp.tool()
def get_price(name: str) -> str:
    """
    Gets the price of a cryptocurrency.
    Args:
        crypto: symbol of the cryptocurrency (e.g., 'bitcoin', 'ethereum').

    ...
```

Live Demo



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Custom MCP Server : API-Calls

Test MCP Servers

Run MCP inspector

```
mcp dev get_crypto_prices.py
```

▶ Connect

Live Demo



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Custom MCP Server : API-Calls

Test MCP Servers

Live Demo

ResourcesPromptsToolsPing# SamplingElicitationsRootsAuthMetadata

Tools

get_price

Gets the price of a cryptocurrency. Args: crypto: symbol of the cryptocurrency (e.g., 'bitcoin', 'ethereum').

get_price

Gets the price of a cryptocurrency. Args: crypto: symbol of the cryptocurrency (e.g., 'bitcoin', 'ethereum').

get_price

Gets the price of a cryptocurrency.

Args:
crypto: symbol of the cryptocurrency (e.g., 'bitcoin', 'ethereum').

name *

arbitrum

Tool-specific Metadata:
No metadata pairs.

Output Schema:

```
{
  type: "object"
  properties: {
    result: {
      type: "string"
    }
  }
}
```

Run Tool

Copy Input

Tool Result: Success

Structured Content:

```
{
  result: "The price of arbitrum is $0.221751 USD."
}
```

Custom MCP Server: API-Calls

Install and Test MCP Servers (in Cursor)

mcp install screenshot.py

```
"Get Crypto Prices": {  
    "command": "uv",  
    "args": [  
        "run",  
        "C:\\\\Projects\\\\LiveTrainings\\\\MCP_Kurzkurs  
        \\\\code\\\\get_crypto_prices.py"  
    ]  
}
```

Live Demo

wie ist der aktuelle preis von arbitrum

Thought for 3s

Ich werde den aktuellen Preis von Arbitrum für Sie abrufen.

✓ Ran get_price: Get Crypto Prices ↕ ...

name arbitrum

Thought for 1s

Der aktuelle Preis von Arbitrum (ARB)
beträgt \$0.221932 USD.

Prompts and Resources

Prompts and Resources

Introduction

Prompts

- Pre-defined templates that guide LLM interactions

Resources

- Structured data for additional context to the LLM

Tools

- Executable functions

For Edge cases

The workhorse



Prompts and Resources

Resources / Tools and Clients

Find out whether MCP Clients use Resources or Tools

Client details

Showing 31 of 103 clients✕ Clear filters

Filter by capabilities:

☒ Resources ☒ Prompts ☐ Tools ☐ Discovery ☐ Sampling ☐ Roots

☐ Elicitation ☐ Instructions

<https://modelcontextprotocol.io/clients>



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Prompts and Resources

Prompts: Script

code > prompts.py >

```
from mcp.server.fastmcp import FastMCP

mcp = FastMCP("Prompt Example")

@mcp.prompt()
def get_prompt(topic: str) -> str:
    """
    Return prompt for doing an analysis on topic {topic}
    Args:
        topic: the topic to analyze
    """
    return f"Perform an analysis on the following topic: {topic}"

if __name__ == "__main__":
    mcp.run()
```

Live Demo



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Prompts and Resources

Prompts: Test MCP Server

Live Demo

Resources Prompts Tools Ping Sampling Elicitations Roots Auth Metadata

Prompts



List Prompts

Clear

get_prompt

Return prompt for doing an analysis on topic {topic} Args: topic: the topic to analyze



get_prompt

Return prompt for doing an analysis on topic {topic} Args: topic: the topic to analyze

topic

Generative AI



Get Prompt

```
{
  description: "
    Return prompt for doing an analysis on topic {topic}
    Args:
      topic: the topic to analyze
  "
  messages: [
    0: {
      role: "user"
      content: { ... } 2 items
    }
  ]
}
```

Prompts and Resources

Resources

- Resources used to retrieve data



MCP Server Deployment

MCP Server Deployment

Npm Packages

- Packages usually created in Typescript/JS and deployed via npm
- We will deploy

Live Demo

npm

The screenshot shows the npm search results for the query 'mcp-server'. The search bar at the top contains 'mcp-server' and a 'Search' button. To the right of the search bar are 'Sign Up' and 'Sign In' buttons. Below the search bar, a list of search results is displayed. Each result includes the package name, a brief description, and the version number. The results are as follows:

Package Name	Description	Version
mcp-server	mcp server	0.0.9
@upstash/context7-mcp	MCP server for Context7	2.1.0
@modelcontextprotocol/server-filesystem	MCP server for filesystem access	2025.12.18
@sap-ux/fiori-mcp-server	SAP Fiori - Model Context Protocol (MCP) server	0.6.7
@notionhq/notion-mcp-server	Official MCP server for Notion API	2.0.0

Below the list of results, there is a link to '4 Versions'.

<https://www.npmjs.com>



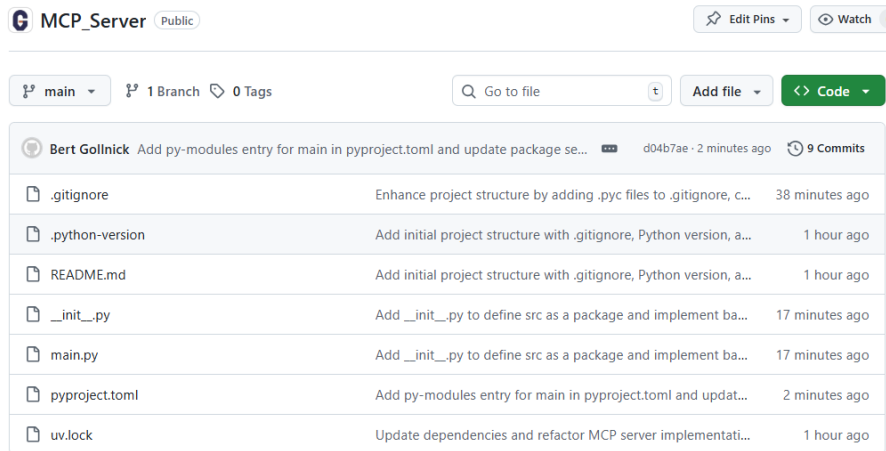
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MCP Server Deployment

Uvx Package development

Live Demo

■ Develop MCP server



The screenshot shows the GitHub repository page for 'MCP_Server' by Bert Gollnick. The repository is public and has 1 branch and 0 tags. The commit history shows 9 commits. The file list includes .gitignore, .python-version, README.md, __init__.py, main.py, pyproject.toml, and uv.lock. The commit messages describe the development of the MCP server, including adding py-modules entry for main in pyproject.toml and updating package structure.

File	Commit Message	Time Ago
.gitignore	Enhance project structure by adding .pyc files to .gitignore, c...	38 minutes ago
.python-version	Add initial project structure with .gitignore, Python version, a...	1 hour ago
README.md	Add initial project structure with .gitignore, Python version, a...	1 hour ago
__init__.py	Add __init__.py to define src as a package and implement ba...	17 minutes ago
main.py	Add __init__.py to define src as a package and implement ba...	17 minutes ago
pyproject.toml	Add py-modules entry for main in pyproject.toml and updat...	2 minutes ago
uv.lock	Update dependencies and refactor MCP server implementati...	1 hour ago



The screenshot shows a live demo interface with a text input field containing 'add'. Below the input field, there is a JSON object representing the request: `{ 'a': 2, 'b': 2 }`. The response is displayed as 'Antwort' with the value '4'. Below the response, it says 'The result of adding 2 and 2 is 4.'

Test

https://github.com/GollnickDataSolutions/MCP_Server



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Summary / Q&A

Summary

f

- s



Contact

My Contact Information



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AI-Training



AI-Consulting



AI-Development



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